

# NAVINSHANKAR V

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## Education

### Northeastern University

*Master of Science in Information Systems, GPA : 3.9*

**September 2023 – December 2025**

*San Jose, California*

### Anna University

*Bachelor of Engineering in Computer Science & Engineering*

**August 2018 – May 2022**

*Chennai, India*

## Experience

### Software Engineer

*TipTop Technologies*

**March 2026 – Present (July 2026)**

*San Jose, California*

- Designed and maintained low-latency Python backend APIs for a real-time communication platform, cutting call-setup latency 25% and sustaining business-critical video and messaging workflows under production load.
- Designed the session-handling layer for LiveKit video rooms to support concurrent room joins without blocking, cutting P99 room-join latency from 480ms to 210ms.
- Built a fault-tolerant reconnect and room-recovery flow that cut dropped-session rate from 3.1% to 0.6%, keeping calls stable through network drops and server restarts.

### Software Engineer Intern

*IpserLab*

**February 2025 – May 2025**

*San Jose, California*

- Built data pipelines feeding ML models from a PostgreSQL store, validating and organizing large datasets to power real-time inference and analytics for 30K+ daily users.
- Built FastAPI backend services following RESTful API design principles, serving ML inference endpoints at sub-200ms response times for 100K+ users, backed by PostgreSQL (RDS) for real-time data workflows.
- Cut configuration drift incidents by 80% through infrastructure design, automating provisioning across 4 environments with Terraform + GitHub Actions.

### Software Developer

*Viatore Solutions*

**October 2022 – July 2023**

*Coimbatore, India*

- Architected an event-driven, distributed serverless platform for real-time device event processing (Lambda, SQS, DynamoDB, IoT Core, S3), applying asynchronous messaging patterns to cut event-processing latency 92%.
- Designed resilient systems improving data accuracy and throughput across 3 business domains at 100% reliability.
- Led Lambda Layer migration applying software design patterns to consolidate 1,000+ lines of duplicated code across functions.
- Orchestrated CI/CD pipelines in GitHub Actions and GitLab CI also enabling same-day hot-fix releases across 3 production microservices and reducing failed deployments 70%.

### Software Developer Intern

*Thirdware Solutions Inc*

**February 2022 – August 2022**

*Chennai, India*

- Built responsive Angular/Bootstrap interfaces for an internal portal serving 2,000+ users, cutting page load time 35%.
- Implemented OAuth 2.0 + JWT authentication with role-based access control, securing API endpoints for 500+ users.

## Projects

### Research Summarizer | *Python, FastAPI, OpenAI API, Render, Docker*

**December 2025**

- Built a RAG summarization service with automated test pipelines validating LLM output on 50+ page inputs, improving reliability through prompt refinement and filters that caught low-confidence and off-topic results.

### HashtagHolidays | *Python, FastAPI, Docker, MongoDB, DialogFlow*

**August 2025**

- Built a FastAPI-based recommendation service using Docker to process social interaction data, applying frequency-recency scoring, persisting ranked results in MongoDB, and exposing them through REST APIs and a conversational interface

### Serpent Struggle | *Python, Pygame*

**August 2025**

- Built a real-time Snake game with four AI-controlled enemy snakes using a greedy path-finding algorithm to compete for food, plus collision detection and progressive difficulty scaling. Implemented the game loop, real-time input handling, and body/head-on collision logic that dynamically resizes or eliminates snakes on impact.

## Technical Skills

**Languages:** Python, Java, JavaScript, SQL, Bash, Linux

**Frameworks:** FastAPI, Django, Spring Boot, Express.js, React, Angular, LiveKit, Bootstrap

**Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis

**Cloud & DevOps:** Docker, Kubernetes, AWS (Lambda, SQS/SNS, ECS, ECR, Fargate, SageMaker, RDS), GCP-ready, Terraform, GitHub Actions, GitLab CI/CD